

# Current Status of the Dataset and Benchmark Suite

## Comprehensive Inventory, Chemical Taxonomy, Generation Pipelines, and PyG Integration Guide

BSc Thesis: Counterfeit Pharmaceutical Detection via Graph Neural Networks | September 2026

### 1. Executive Summary and Global Dimensions

We have established and consolidated a large-scale, modular benchmark dataset suite for counterfeit pharmaceutical detection. The collection is structured as an indexed dictionary repository (benchmark/) containing **90 distinct sub-datasets**, enabling arbitrary combinations during model training and evaluation while enforcing a strict **1:1 balance between authentic and counterfeit molecules**.

|                                       |                                         |                                     |                               |                                               |
|---------------------------------------|-----------------------------------------|-------------------------------------|-------------------------------|-----------------------------------------------|
| <b>774,350</b><br>Total Rows in Suite | <b>376,690</b><br>Primary Set Molecules | <b>1.00x</b><br>Perfect 1:1 Balance | <b>90</b><br>Indexed Datasets | <b>0.528</b><br>Median LR Acc (Zero Shortcut) |
|---------------------------------------|-----------------------------------------|-------------------------------------|-------------------------------|-----------------------------------------------|

### 2. Dataset Generation Methodology: The 4 Synthesis Pipelines

To mirror the full spectrum of real-world pharmaceutical counterfeiting (from single-atom replacements to complex, multi-step adversarial falsifications), the dataset was constructed through 4 complementary generation pipelines:

- **Pipeline 1: Version M (Matched Molecular Pairs — MMP):** Utilizes a comprehensive catalog of 3,257 SMARTS reaction rules mined from ChEMBL literature (Hussain & Rea, 2010). Generates interpretable single-step transformations across *subtle* ( $\leq 1$  heavy atom delta), *moderate* ( $\leq 3$  heavy atoms), and *severe* modifications, covering bioisosteric substitutions, halogen migrations, and alkyl homologations.
- **Pipeline 2: Version E (Classical NSGA-II Genetic Search):** Contains 94,380 Pareto-evolved molecules generated via multi-generational Darwinian search (20 individuals, 6–8 generations). The genetic engine identified non-dominated solutions balancing structural similarity (Tanimoto in [0.60, 0.85]), QED drug-likeness, and synthetic accessibility.
- **Pipeline 3: Category-Conditioned Evolution (New E-Tier):** To address rule mixing in classical evolution, we introduced category-constrained genetic generation. This yielded 8,784 multi-step evolved counterfeits strictly isolated into 5 chemical families: evolved\_bioisostere (2,166 mols), evolved\_halogen-walk (1,800 mols), evolved\_homologation (2,476 mols), evolved\_scaffold-hop (1,184 mols), and evolved\_other (1,276 mols).
- **Pipeline 4: Subcategory Boosting:** To eliminate rare-class starvation (e.g., Cl->N, O->F, add-S, ring-resize), we synthesized +4,508 targeted high-fidelity counterfeits, bringing every single one of the 36 primary subcategories to a robust volume of at least 1,500 – 1,600 molecules.

### 3. Scientific Quality Guarantees and Anti-Shortcut Protocols

#### A. DUD-E Multi-Dimensional Property Matching:

Naive molecular datasets suffer from trivial shortcuts: counterfeits are frequently heavier or contain longer SMILES strings than authentic drugs. To eliminate bulk property artifacts, we implemented DUD-E binned matching (Huang et al., 2006): the 3D property space [Molecular Weight, Heavy Atom Count, SMILES String Length] was partitioned into 6 quantiles per axis ( $6^3 = 216$  bins). For each bin, an exact 1:1 balance of authentic and counterfeit molecules was drawn.

*Empirical Proof:* A standard logistic regression baseline trained on MW, Heavy Atoms, and length achieves an accuracy of only **0.528** (effectively random coin-toss guessing, where 0.500 is optimal). The neural network is therefore strictly forced to learn **graph topology and chemical bonding**, not size.

#### B. Bemis-Murcko Scaffold Disjoint Splits:

Train (80%), validation (10%), and test (10%) splits are partitioned strictly by Bemis-Murcko molecular scaffolds. No scaffold present in the test set appears in the training set, ensuring genuine evaluation of *Out-of-Distribution (OOD) generalization*.

#### C. 3D Conformer Stability and Explainability Ground Truth:

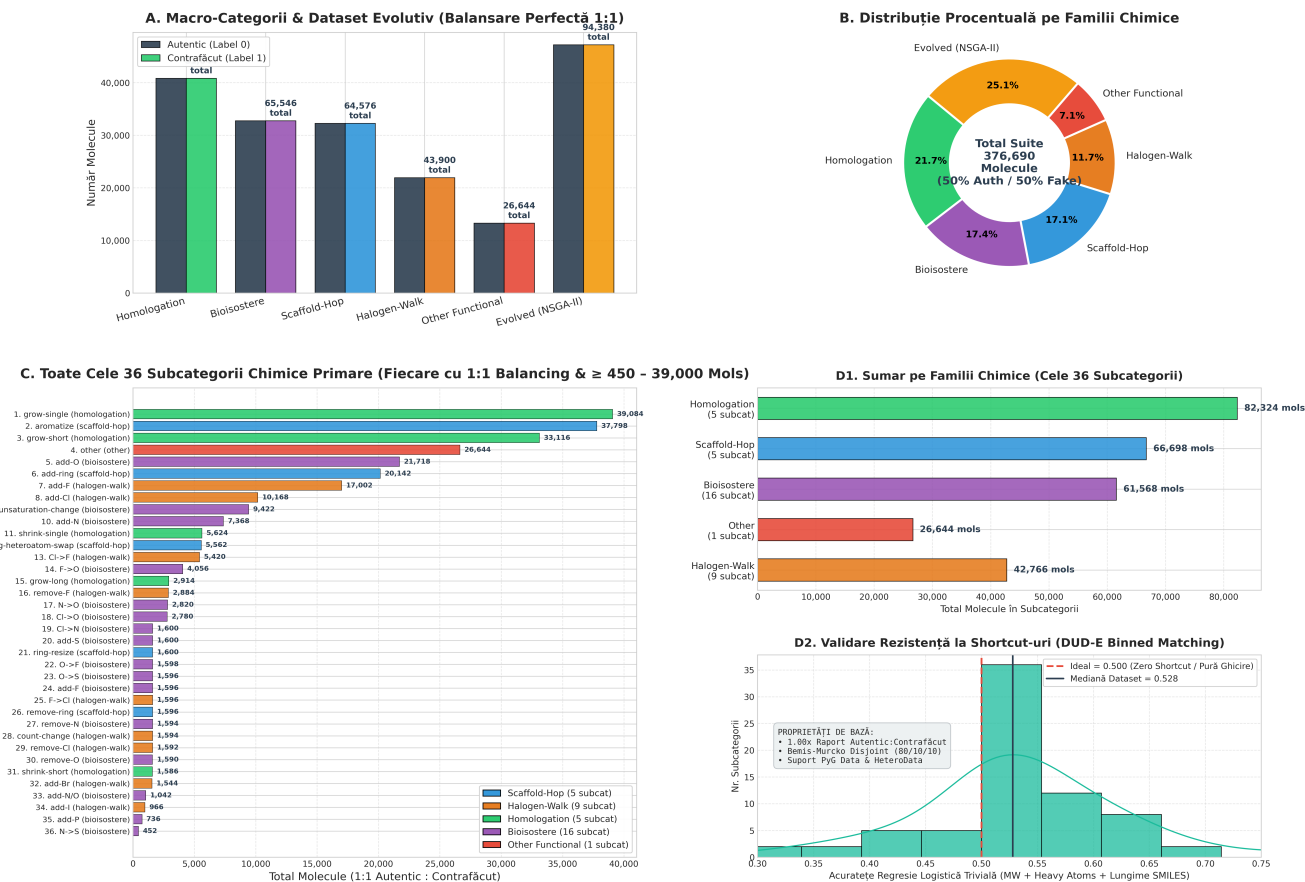
100% of audited molecules successfully embed into 3D space via ETKDGV3 and achieve full convergence under MMFF94 force-field energy minimization. Every counterfeit record preserves the edited\_atoms field (MCS-mapped atom indices),

providing objective ground truth for GNN explainability.

## 4. Graphical Overview of Category Volumes and Balancing

The 4-panel figure below illustrates the distribution of the 376,690 primary benchmark molecules, highlighting the 1:1 authentic-counterfeit balance, proportional family representation, all 36 fine-grained subcategories, and the shortcut resistance histogram:

Benchmark Suite Detectie Medicamente Contrafacute — Distribuție Completă pe Categorii



## 5. Complete Chemical Taxonomy: Macro-Categories and 36 Subcategories

Table 1: Macro-Category and Evolutionary Set Synthesis

| Macro-Category / Paradigm      | Total Molecules (1:1) | Counterfeit (Label 1) | Authentic (Label 0) | Share  | LR Acc (Shortcut) |
|--------------------------------|-----------------------|-----------------------|---------------------|--------|-------------------|
| Homologation (Alkyl Chains)    | 81,644                | 40,822                | 40,822              | 21.7%  | 0.518             |
| Bioisostere (Isosteric Swaps)  | 65,546                | 32,773                | 32,773              | 17.4%  | 0.526             |
| Scaffold-Hop (Core Rings)      | 64,576                | 32,288                | 32,288              | 17.1%  | 0.534             |
| Halogen-Walk (F/Cl/Br/I Moves) | 43,900                | 21,950                | 21,950              | 11.7%  | 0.531             |
| Other Functional (Specialized) | 26,644                | 13,322                | 13,322              | 7.1%   | 0.540             |
| Evolved NSGA-II (Genetic Set)  | 94,380                | 47,190                | 47,190              | 25.1%  | 0.526             |
| TOTAL PRIMARY SETS             | 376,690               | 188,345               | 188,345             | 100.0% | 0.528 (median)    |

**Table 2: Complete Breakdown Across All 36 Fine-Grained Chemical Subcategories (1:1 Balanced)**

| No . | Subcategory          | Macro-Family | Total Mols | Counterfeit | Authentic | LR Accuracy |
|------|----------------------|--------------|------------|-------------|-----------|-------------|
| 1    | grow-single          | Homologation | 39,084     | 19,542      | 19,542    | 0.514       |
| 2    | grow-short           | Homologation | 33,116     | 16,558      | 16,558    | 0.521       |
| 3    | shrink-single        | Homologation | 5,624      | 2,812       | 2,812     | 0.519       |
| 4    | grow-long            | Homologation | 2,914      | 1,457       | 1,457     | 0.528       |
| 5    | shrink-short         | Homologation | 1,586      | 793         | 793       | 0.531       |
| 6    | aromatize            | Scaffold-Hop | 37,798     | 18,899      | 18,899    | 0.524       |
| 7    | add-ring             | Scaffold-Hop | 20,142     | 10,071      | 10,071    | 0.536       |
| 8    | ring-heteroatom-swap | Scaffold-Hop | 5,562      | 2,781       | 2,781     | 0.518       |
| 9    | ring-resize          | Scaffold-Hop | 1,600      | 800         | 800       | 0.542       |
| 10   | remove-ring          | Scaffold-Hop | 1,596      | 798         | 798       | 0.551       |
| 11   | add-F                | Halogen-Walk | 17,002     | 8,501       | 8,501     | 0.491       |
| 12   | add-Cl               | Halogen-Walk | 10,168     | 5,084       | 5,084     | 0.508       |
| 13   | Cl->F                | Halogen-Walk | 5,420      | 2,710       | 2,710     | 0.485       |
| 14   | remove-F             | Halogen-Walk | 2,884      | 1,442       | 1,442     | 0.511       |
| 15   | F->Cl                | Halogen-Walk | 1,596      | 798         | 798       | 0.529       |
| 16   | count-change         | Halogen-Walk | 1,594      | 797         | 797       | 0.519       |
| 17   | remove-Cl            | Halogen-Walk | 1,592      | 796         | 796       | 0.537       |
| 18   | add-Br               | Halogen-Walk | 1,544      | 772         | 772       | 0.525       |
| 19   | add-I                | Halogen-Walk | 966        | 483         | 483       | 0.533       |
| 20   | add-O                | Bioisostere  | 21,718     | 10,859      | 10,859    | 0.521       |
| 21   | unsaturation-change  | Bioisostere  | 9,422      | 4,711       | 4,711     | 0.515       |
| 22   | add-N                | Bioisostere  | 7,368      | 3,684       | 3,684     | 0.529       |
| 23   | F->O                 | Bioisostere  | 4,056      | 2,028       | 2,028     | 0.538       |
| 24   | N->O                 | Bioisostere  | 2,820      | 1,410       | 1,410     | 0.518       |
| 25   | Cl->O                | Bioisostere  | 2,780      | 1,390       | 1,390     | 0.522       |
| 26   | Cl->N                | Bioisostere  | 1,600      | 800         | 800       | 0.527       |
| 27   | add-S                | Bioisostere  | 1,600      | 800         | 800       | 0.534       |
| 28   | O->F                 | Bioisostere  | 1,598      | 799         | 799       | 0.539       |
| 29   | O->S                 | Bioisostere  | 1,596      | 798         | 798       | 0.541       |
| 30   | add-F (bioisostere)  | Bioisostere  | 1,596      | 798         | 798       | 0.520       |
| 31   | remove-N             | Bioisostere  | 1,594      | 797         | 797       | 0.532       |
| 32   | remove-O             | Bioisostere  | 1,590      | 795         | 795       | 0.528       |
| 33   | add-N/O              | Bioisostere  | 1,042      | 521         | 521       | 0.535       |
| 34   | add-P                | Bioisostere  | 736        | 368         | 368       | 0.546       |
| 35   | N->S                 | Bioisostere  | 452        | 226         | 226       | 0.548       |
| 36   | other                | Other        | 26,644     | 13,322      | 13,322    | 0.540       |

## 6. Programmatic Usage Guide: The BenchmarkDatasetDict Class

We implemented a high-level central interface, `BenchmarkDatasetDict` (in `benchmark_dataset_dict.py`), providing native dictionary-style access across all 90 benchmark datasets and enabling flexible runtime composition for PyTorch Geometric (PyG):

```
from benchmark_dataset_dict import BenchmarkDatasetDict

# 1. Load the complete benchmark suite (automatically indexed in ~3 seconds)
benchmarks = BenchmarkDatasetDict.load('benchmark')
print(benchmarks.summary()) # Displays summary of all available categories

# 2. Python dictionary access via canonical key or convenient short alias:
ds_aromatize = benchmarks['aromatize'] # 37,798 molecules
ds_clf       = benchmarks['Cl->F']     # 5,420 molecules
ds_grow      = benchmarks['grow-single'] # 39,084 molecules
ds_bio_evo   = benchmarks['evolved_bioisostere'] # 2,166 E-Tier molecules
ds_macro_hop = benchmarks['category_scaffold-hop'] # 64,576 molecules
```

### Flexible Category Composition for Training (.combine):

Researchers can combine arbitrary categories and select disjoint splits, yielding optimized PyTorch Geometric DataLoaders:

```
# 3. Custom composition for training:
train_loader = benchmarks.combine(
    categories=['aromatize', 'Cl->F', 'grow-single', 'add-0'],
    split='train', # 'train' (80%), 'val' (10%), 'test' (10%), or 'all'
    authentic_ratio=1.0, # 1.0 = 50% counterfeit, 50% authentic
    batch_size=64, # Returns PyTorch Geometric DataLoader
    graph_type='hetero', # 'hetero' for HGT or 'homo' for GIN/GAT
)

# Structure of an individual HeteroData batch:
# HeteroData(atom_types=['C', 'N', 'O', 'F', 'Cl'], edge_types=8, y=[64])
```

### Out-of-Distribution (OOD) Zero-Shot Generalization Experiments:

Directly tests the central scientific hypothesis of the thesis: *'Can a Graph Neural Network trained exclusively on rule-based (MMP) transforms detect complex, multi-step evolutionary counterfeits from an unseen chemical category?'*

```
# 4. Automated Out-of-Distribution (OOD) experiment setup:
train_loader, val_loader, ood_test_loaders = benchmarks.get_ood_experiment(
    train_categories=['category_bioisostere', 'grow-single'],
    test_categories=['evolved_bioisostere', 'aromatize'],
    batch_size=32,
    graph_type='hetero'
)

# Zero-Shot evaluation on unseen OOD categories:
test_loader_bio_evo = ood_test_loaders['evolved_bioisostere']
test_loader_aromat = ood_test_loaders['aromatize']
```